

Evanns Morales-Cuadrado

U.S. Citizen

Robotics PhD Candidate Georgia Institute of Technology

CONTACT

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Github: <https://github.com/evannsmc>

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, C#/.NET, Matlab/Simulink
- **Robotics & Control Frameworks:** ROS 2, NATS, PX4, ArduPilot, acados, CasADi
- **Machine Learning & Simulation:** JAX, PyTorch, MuJoCo, Reinforcement Learning
- **Development & DevOps Tools:** Docker, Continuous Integration, Git/GitHub
- **Hardware & Experimentation:** UAV assembly, repair, and deployment
- **Motion-Capture:** OptiTrack and Vicon
- **Languages:** Spanish (Native), English (Native)

EDUCATION

Georgia Institute of Technology | 2022-2027

Ph.D. in Robotics

Doctoral Minor in Optimization Methods

- GPA: 4.0/4.0
- Advisor: Dr. Samuel Coogan
- Thesis Proposal Approved (ABD)
- Goizueta Fellow

University of Texas at Arlington | 2018-2022

Honors Bachelor of Science in Electrical Engineering

Minor in Mathematics

Minor in Physics

Certificate in Unmanned Vehicle Systems

- GPA: 3.9/4.0 Summa Cum laude
- Full National Merit Scholarship Awarded
- Electrical Engineering Honors Scholar

WORK EXPERIENCE

Graduate Research Intern | AI for Autonomy

Sandia National Laboratories

May 2026 - Present

Albuquerque, NM

- Conducted applied research using reinforcement learning for autonomous systems
- Developed C#/.NET networking frameworks and execution scripts for coordinating communication and task execution across multi-robot systems
- Created novel robotics and automation infrastructure to advance national interests

Teaching Assistant | Vertically Integrated Project in Robotics

Georgia Institute of Technology

January 2024 - Present

Atlanta, GA

- Aided professor in his instruction of this research-based course
- Provided promising undergraduates with guidance performing applied research in the field of robotics
- Taught control theoretical concepts in an accessible manner for undergraduates

Graduate Research Assistant | FACTS Lab

Georgia Institute of Technology

August 2022 - Present

Atlanta, GA

- Advised by Dr. Samuel Coogan
- Implemented novel aggressive tracking control methods on quadrotor hardware
- Developed novel nonlinear control methods for safe autonomy
- Published novel research at top conferences and journals in the fields of robotics and control theory

Research Assistant | Autonomous Systems Lab

University of Texas at Arlington

November 2021 - May 2022

Arlington, TX

- Advised by Dr. Frank Lewis
- Received personal mentorship from a leading researcher in the field of controls and reinforcement learning (ranked 12th in the USA and 23rd in the world during my time in his lab)
- Aided graduate students on reinforcement learning research applied to quadrotor control

Teaching Assistant | Graduate-Level Intelligent Systems Course

University of Texas at Arlington

January 2022 - May 2022

Arlington, TX

- Aided professor in his instruction of the course
- Hosted review sessions for students in preparation for exams
- Gained teaching experience in a graduate-level course while still an undergraduate

Research Assistant | Dynamical Networks and Control Lab

University of Texas at Arlington

October 2019 - May 2022

Arlington, TX

- Advised by Dr. Yan Wan
- Gained experience with Robot Operating System (ROS), OpenCV, and path planning
- Original research in learning-based minimum time and energy path-planning for multi-vehicle systems

SELECTED PUBLICATIONS

M. Cao, A. Harapanahalli, **E. Morales-Cuadrado**, and S. Coogan, “Trajectory Tracking for Systems with Unknown Time-Varying Disturbances.” Submitted to Open Journal of Control Systems. (*under revision*)

E. Morales-Cuadrado, “Computationally Efficient and Safe Control for Aerial Robotic Systems under Threat and Disturbance.” Ph.D. Thesis Proposal, Georgia Institute of Technology, 2026. [pdf available](#). [slides available](#).

E. Morales-Cuadrado, L. Chung, S. Kousik and S. Coogan, “RTD-RAX: Fast, Safe Trajectory Planning for Systems under Unknown Disturbances.” Joint Submission to 2026 Modeling, Estimation and Control Conference (MECC) and ASME Letters in Dynamic Systems and Control (ALDSC). (*under revision*) [preprint available](#).

E. Morales-Cuadrado, L. Baird, Y. Wardi and S. Coogan, “Lightweight Tracking Control for Computationally Constrained Aerial Systems with the Newton-Raphson Method.” *IEEE Transactions on Control Systems Technology*, 2026. (*early access; to appear*) [preprint available](#).

L. Baird, **E. Morales-Cuadrado**, and S. Coogan, “Runtime Assurance for Uncertain Systems from Interval Signal Temporal Logic.” Submitted to IFAC Nonlinear Analysis: Hybrid Systems (NAHS), 2025. (*under revision*) [preprint available](#).

E. Morales-Cuadrado, C. Llanes, Y. Wardi and S. Coogan, “Newton-Raphson Flow for Aggressive Quadrotor Tracking Control.” *2024 American Control Conference (ACC)*

RESEARCH INTERESTS

- | | |
|---------------------------------------|--------------------------------------|
| • Safe Autonomy | • Advanced Nonlinear Control |
| • Hardware Deployment | • Learning-Based Control |
| • Unmanned Ground and Aerial Vehicles | • Trajectory Generation and Planning |

OPEN-SOURCE CONTRIBUTIONS

- **PX4 Autopilot:** Contributed documentation improvements to the official PX4 Autopilot manual.
- **Vicon4PX4:** Developed and open-sourced a ROS 2 C++ Vicon–PX4 interface enabling external vision fusion for indoor quadrotor flight experiments; adopted and showcased by Georgia Tech’s Indoor Flight Lab as part of its official codebase. Also created a parallel package for OptiTrack systems.

SERVICE & LEADERSHIP

Co-Founder and President of Puerto Rican Student Association Georgia Institute of Technology	August 2024 - Present Atlanta, GA
• Addressed the need for an organization to help Puerto Rican students feel at home and stay connected to our culture • Formed a three-member co-founder board and identified a faculty advisor • Raised funds for the organization and recruited members from campus • Hosted professional, social, and cultural events to meet the needs of the growing Puerto Rican student population at Georgia Tech	
Undergraduate Research Mentor University of Texas at Arlington	August 2021 - May 2022 Arlington, TX
• As the lead undergraduate student Dr. Frank Lewis' Autonomous Systems Lab, I mentored junior undergrads in controls research and guided them on a research project in ground vehicle control • Guided students in Robot Operating System, Python, Linux, and Control Systems	
Teaching Assistant Graduate-Level Intelligent Systems Course University of Texas at Arlington	January 2022 - May 2022 Arlington, TX
• Aided professor in his instruction of the course • Hosted review sessions for students in preparation for exams • Graded homeworks, exams, and projects • Gained teaching experience in a graduate-level course while still an undergraduate	
Eta Kappa Nu Public Relations Chair & Circuits I Tutor University of Texas at Arlington	August 2020 - May 2022 Arlington, TX
• Executive board member of UT Arlington's IEEE-HKN chapter • In charge of social media advertising for honor society's events and outreach initiatives for the electrical engineering student body • Volunteered to provide free tutoring as a service to electrical engineering students in Circuits I	

HONORS AND AWARDS

Top-3 Finalist at a Deep Learning Research Symposium Professor-sponsored award for novel research in deep learning and robotics	November 2023
Goizueta Fellowship at Georgia Tech Highly selective fellowship for graduate students of Latin-American descent	August 2022
Summa Cum Laude Honors at University of Texas at Arlington Exceptional academic distinction in top GPA tier for graduation honors	May 2022
Honors College Degree Requirements at University of Texas at Arlington Completed research projects in disparate fields throughout my undergraduate career with various professors and defended a senior capstone research project before the Honors College committee	December 2021
Electrical Engineering Honors Scholar at University of Texas at Arlington Inaugural member of the electrical engineering honors cohort	December 2021
IEEE-HKN International Electrical Engineering Honor Society Membership Selective undergraduate society recognizing outstanding academic achievement and community service	August 2021
Research Experience for Undergraduates Fellowship Recipient Fellowship to sponsor promising young undergraduate researchers	August 2020
Chance Vought Engineering and Science Endowment Scholarship Highly selective award given annually to three recipients across the entire College of Engineering	August 2020
Toray Composite Materials America Inc. Sons and Daughters Scholarship Academic scholarship awarded children of employees of Toray America	May 2019
James E. Casey Scholarship Recipient Academic scholarship awarded children of employees of UPS	January 2019
James R. Hoffa Memorial Scholarship Recipient Academic scholarship awarded children of Teamsters Union Members	May 2018

Grapevine, TX Rotary Club Scholarship Recipient

May 2018

Local rotary club scholarship for academic and civically-minded scholars

National Hispanic Scholar

March 2018

Recognition of top national performers of Latin descent on the SAT

National Merit Scholar

March 2018

Recognition of top national performers on the SAT